

Week-2 EDA Report
Data Visualization Internship



Submitted By:

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Internship Program

1 INTRODUCTION

The outreach and application datasets provide critical information on how prospective students interact with the university's admission process. The **Outreach Campaign Data** records communication attempts with applicants, while the **Application Data** provides demographic and contact details of those applicants. Conducting Exploratory Data Analysis (EDA) allows us to uncover patterns, trends, and anomalies that directly affect the effectiveness of outreach campaigns and the efficiency of the application pipeline.

1.1 Report Overview

This report presents the results of the **Exploratory Data Analysis (EDA)** conducted on two cleaned datasets:

1. **Outreach Campaign Data (37,435 records, 8 columns)**
 - Contains information about outreach calls made to applicants, including call outcomes, caller names, escalation requirements, and timestamps.
 - Used to evaluate outreach effectiveness, caller performance, and applicant engagement patterns.
2. **Application Data (21,088 records, 5 columns)**
 - Captures applicant-level details such as Applicant ID, Country, University Name, Phone, and Email.
 - Used to understand applicant demographics, data completeness, and identify anomalies such as duplicates or missing values.

The analysis is structured into:

- **Dataset Overview** – A description of data size, columns, and missing values.
- **Key Trends** – Insights on outreach performance, escalation rates, applicant demographics, and engagement patterns.
- **Anomalies** – Identification of data inconsistencies such as missing emails, duplicate records, and unusual call outcomes.
- **Visualizations & Tables** – Charts and tabular summaries that illustrate key findings.
- **Insights & Opportunities** – Explanation of why these findings matter and how they align with outreach performance goals.

1.2 Objectives

The objectives of this EDA are to:

1. Provide a structured overview of both datasets (size, structure, missing data).
2. Identify key trends in outreach performance (e.g., most common call outcomes, escalation needs, and caller performance).

3. Highlight anomalies or data quality issues, such as duplicates or missing contact details in the application dataset.
4. Explore segment-level insights (e.g., applicant distribution by country, response patterns over time).
5. Generate actionable recommendations that can guide future outreach campaigns and application follow-ups.

2 DATASET OVERVIEW & EDA

2.1 Outreach Data

Total Records: 37,435

Columns: 8

Time Range: January 2023 – May 2023

Columns Description:

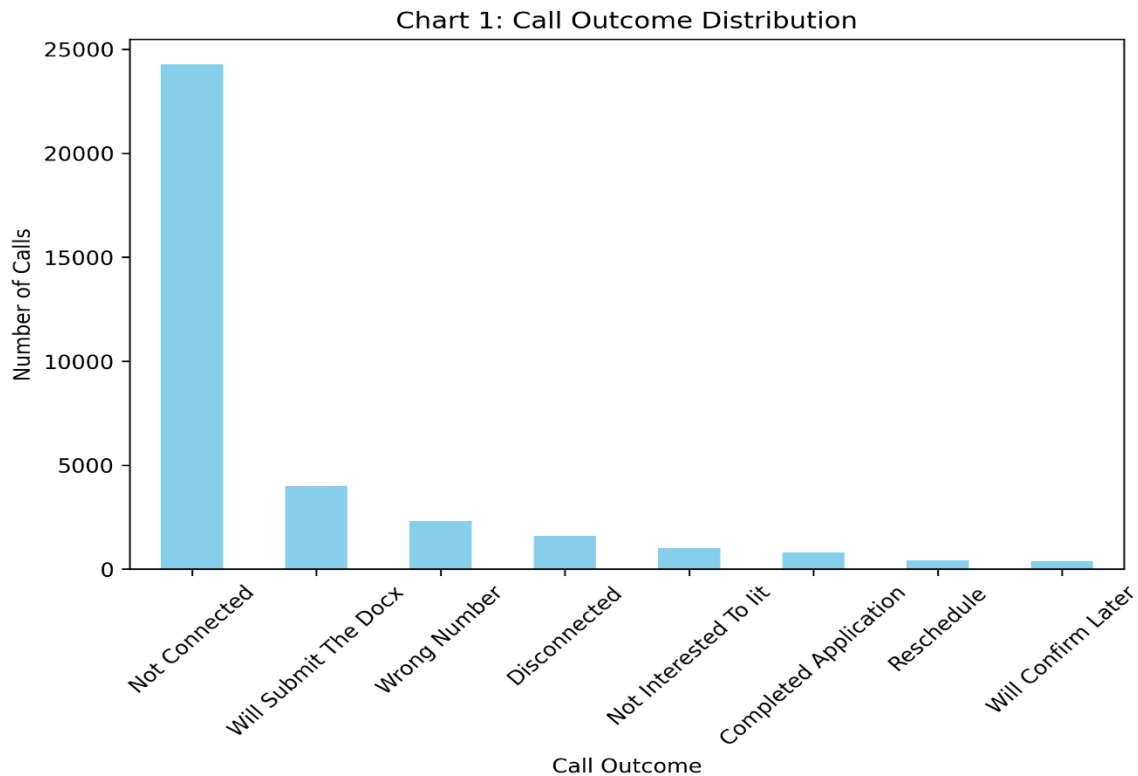
Column Name	Description
Applicant_ID	Unique identifier for each applicant
Received_At	Timestamp of the outreach call
University_Name	University name (Illinois Institute of Technology in this case)
Caller_Name	Name of the caller/agent handling outreach
Call_Outcome	Result of the call (e.g., Connected, Not Connected, Wrong Number, etc.)
Remarks	Additional notes from the caller
Campaign_ID	Identifier for the campaign
Escalation_Required	Whether the call required escalation (Yes/No)

Key Trends & Patterns

Call Outcomes Distribution (Top 10)

Call Outcome	Count
Not Connected	24,259
Will Submit The Docx	3,997
Wrong Number	2,314

Disconnected	1,587
Not Interested To IIT	1,024
Completed Application	813
Reschedule	431
Will Confirm Later	384
Ready To Pay The Deposit	344
Student Has Info & Plans To Enroll Soon	327

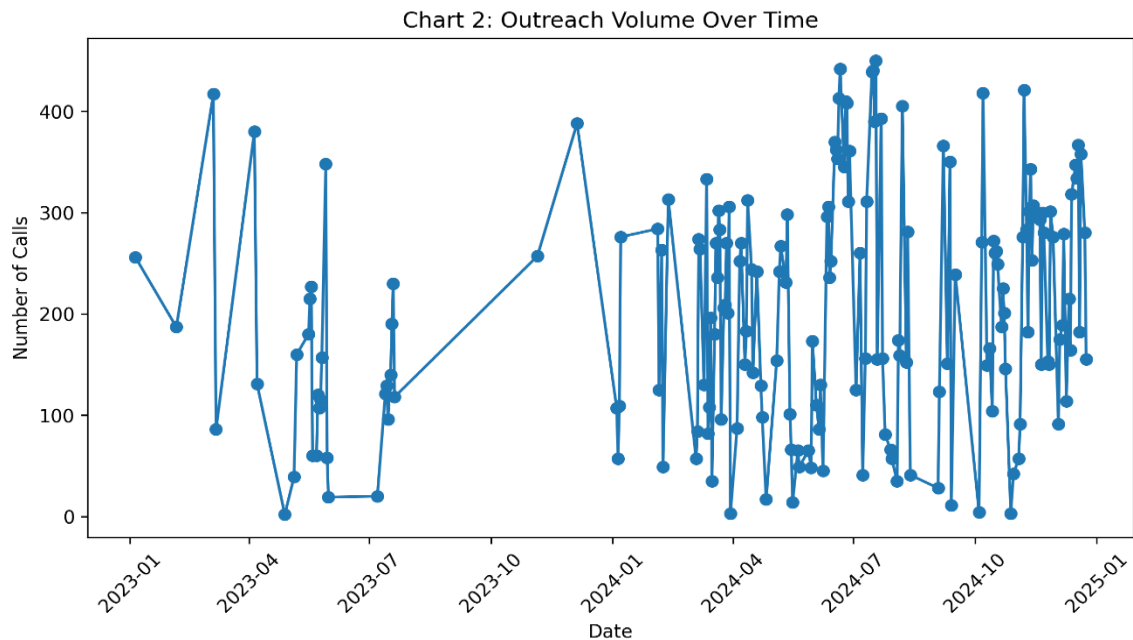


To visualize the balance between positive and negative outcomes.

What it shows: Most outcomes are negative or neutral (Not Connected, Wrong Number), with only a small fraction showing conversion intent. A visual makes it clear how skewed outcomes are and helps stakeholders quickly grasp campaign effectiveness.

Outreach Volume by Date

Date	Calls
2023-01-05	256
2023-02-05	187
2023-03-05	417
2023-03-07	86
2023-04-05	380
2023-04-07	131
2023-04-28	2
2023-05-05	39
2023-05-07	160
2023-05-16	180



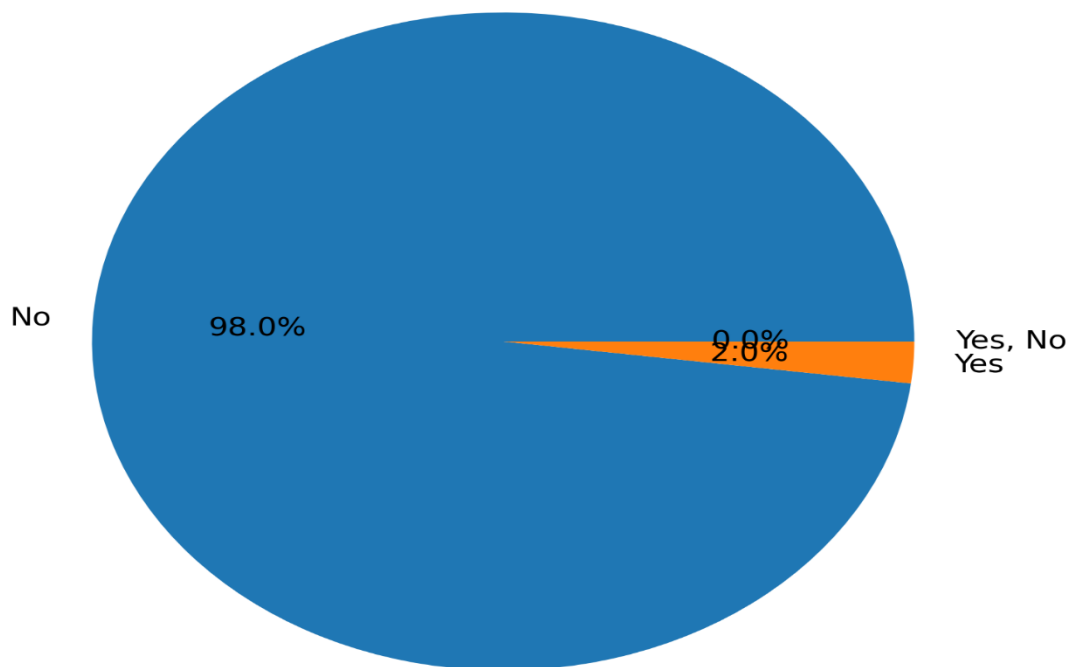
To observe the overall trend of outreach calls aggregated across dates.

Insight: The line shows clear spikes in **March and April** (admission cycle months), followed by lower volumes in **May** and earlier in the year. Helps identify seasonal patterns in outreach campaigns, suggesting the need to spread calls more evenly or intensify efforts during proven peak months

Escalation Required (%)

Escalation Required	%
No	97.96
Yes	2.04
Yes, No	0.003

Chart 3: Escalation Required



To measure how many calls required escalation for further action.

Insight: Almost 98% of calls did **not** require escalation, while only ~2% needed additional support. Shows that most cases are straightforward, but also raises the question of whether complex issues are being under-reported or under-addressed

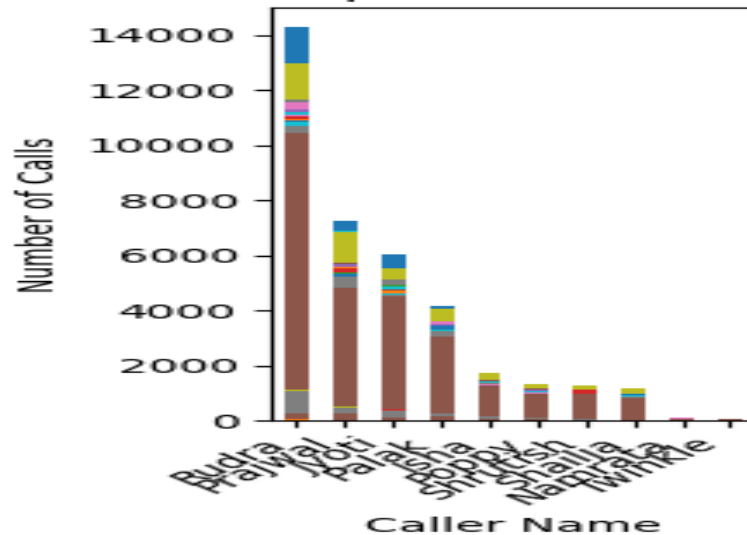
Low escalation rates suggest most calls are routine, but it also raises a question: are complex cases being underreported? Monitoring escalation ensures applicants with issues are supported properly.

Key Anomalies

Caller Performance Comparison

Caller Name	Completed App	Connected	Disconnected	Will Submit Docs	Wrong Number
Isha	54	1	76	233	2
Jyoti	86	0	249	401	478
Palak	118	0	113	432	122
Poppy	38	0	44	189	4
Prajwal	245	0	218	1111	362
Rudra	198	0	821	1302	1300
Shailja	50	2	14	177	4

Caller Performance (Top 10 callers) —



To evaluate the performance of different callers and their outcomes.

What it shows: Some callers (e.g., Rudra, Prajwal) handle large volumes with varied results, while others (e.g., Mounika, Namrata) have very few records.

Why it matters: This identifies high-performing callers and possible training gaps. It also helps allocate calling tasks more effectively.

Summary Outreach Data

Aspect	Details
Total Records	37,435
Top Outcomes	Not Connected (65%), Will Submit Documents (~11%), Wrong Number (~6%)
Positive Outcomes	Completed Application (2%), Ready to Pay Deposit (1%)
Escalations	~2% of calls required escalation
Trend Over Time	Peak outreach activity observed in March–April 2023
Caller Performance	Significant variation; few callers generate most successful outcomes
Key Insight	High “Not Connected” rate → inefficiency; caller-level training needed

2.2 Applicant Dataset

Total Records: 21,088

Columns: 5

Columns Description:

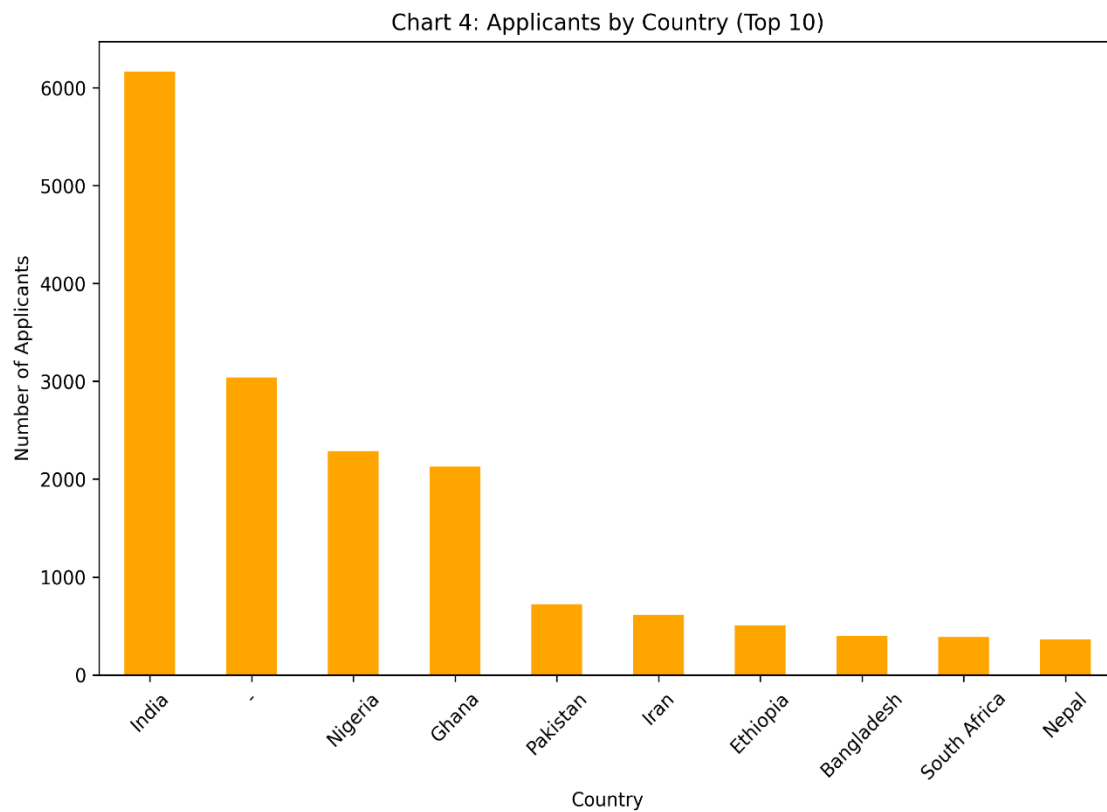
Column Name	Description
Applicant_ID	Unique identifier for each applicant
Country	Country of origin of the applicant
University_Name	University applied to (Illinois Institute of Technology for all records)
Phone	Contact number of applicant
Email	Email address of applicant (very limited availability)

Key Trends & Patterns

Applicants by Country (Top 15)

Country	Count
India	6,167
(missing)	3,039
Nigeria	2,285
Ghana	2,129
Pakistan	720
Iran	613
Ethiopia	505
Bangladesh	398
South Africa	387

Nepal	364
Kenya	279
Morocco	234
USA	183
Botswana	175



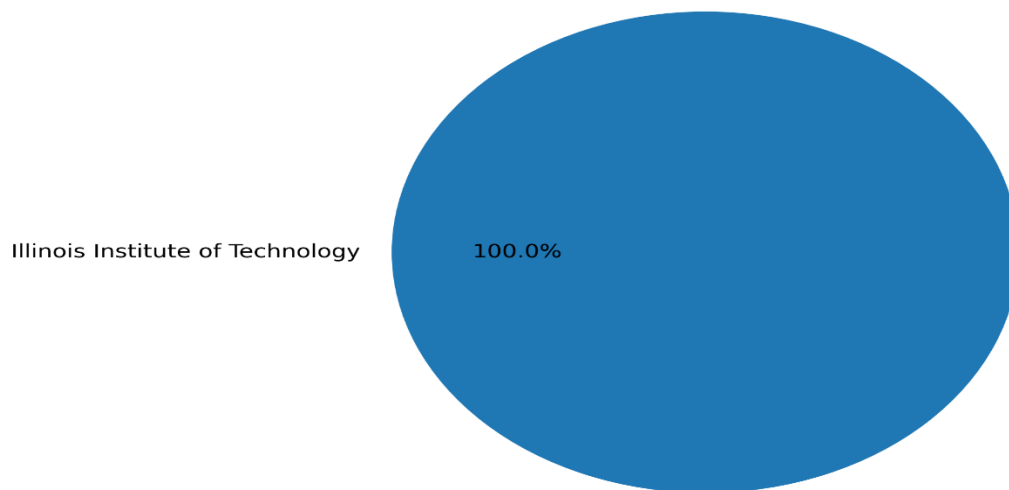
To identify the geographic distribution of applicants.

What it shows: India (6,167), Nigeria (2,285), and Ghana (2,129) dominate applications, while others like Bahrain and Botswana have smaller counts. Helps prioritize outreach campaigns for high-volume countries and customize strategies by region.

University Distribution

University	Count
Illinois Institute of Technology	21,088

Chart 5: University Distribution



To confirm whether applicants are applying to a single institution.

What it shows: All 21,088 applications belong to **Illinois Institute of Technology**. Ensures data relevance—our outreach is focused solely on one university’s pipeline.

Key Anomalies

Missing Contact Information

Field	Missing Count
Phone	0
Email	20,471

The missing values shows in the email column because during cleaning process we found emails in the country column so for that we create an other column named as “ Email” and move that email rows to this column, so we fill missing values with the NA

Summary Applicant Data

Aspect	Details
Total Records	21,088
Country Distribution	India (29%), Nigeria, Ghana, Pakistan, Iran (Top contributors)
Missing Data	Country missing in ~14% records; Email missing in 97%
University	100% applicants for Illinois Institute of Technology
Contact Information	Phone available for all applicants; Email largely missing
Key Insight	Diverse applicant base, but missing emails and duplicates reduce efficiency

2.3 Campaing Dataset

Total Records: 23 unique campaigns

Columns: 7 (Campaign_ID, Campaign_Name, Campaign_Category, Intake, University_Name, Campaign_Status, Start_Date)

Missing Values: None (all fields are complete)

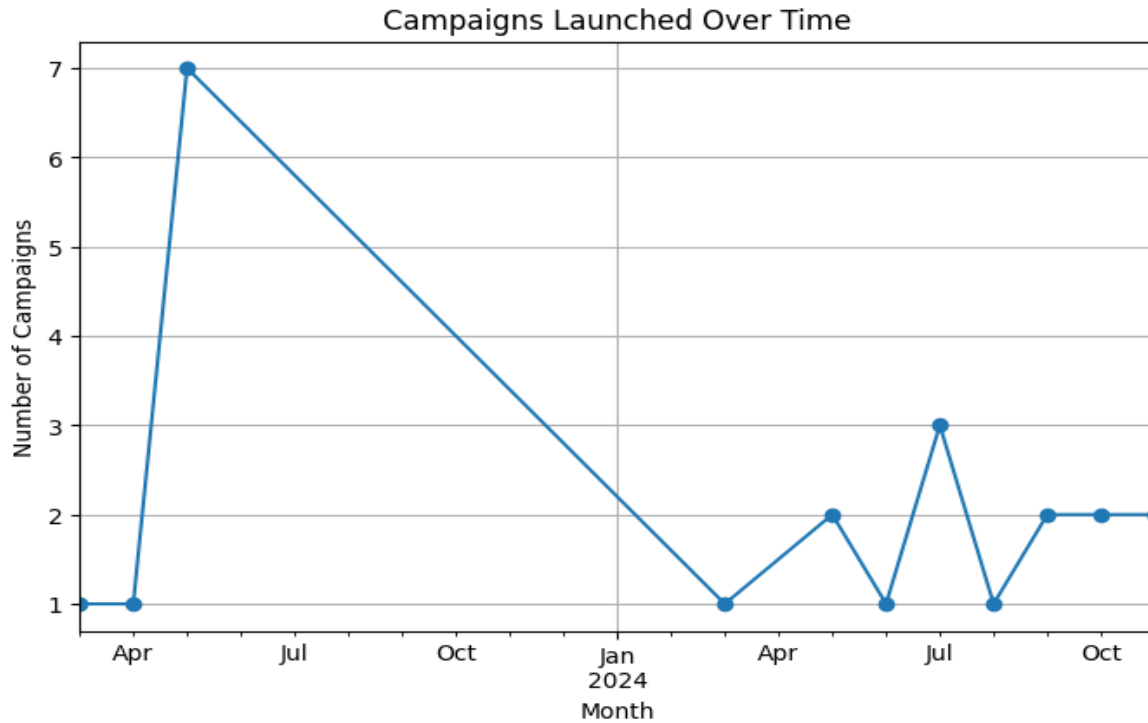
Coverage: Campaigns span across 2023–2024 intakes

Note: The dataset was **linked with the Outreach dataset using Campaign_ID** to align campaign-level details (start date, category, intake, university) with outreach performance (calls, outcomes, escalations). This integration enables cross-analysis of campaign design vs. actual execution.

Key Trends & Patterns

Campaigns Launched by Month

Month	Campaigns Launched
2023-03	1
2023-04	1
2023-05	7
2024-03	1
2024-05	2
2024-06	1
2024-07	3
2024-08	1
2024-09	2
2024-10	2
2024-11	2

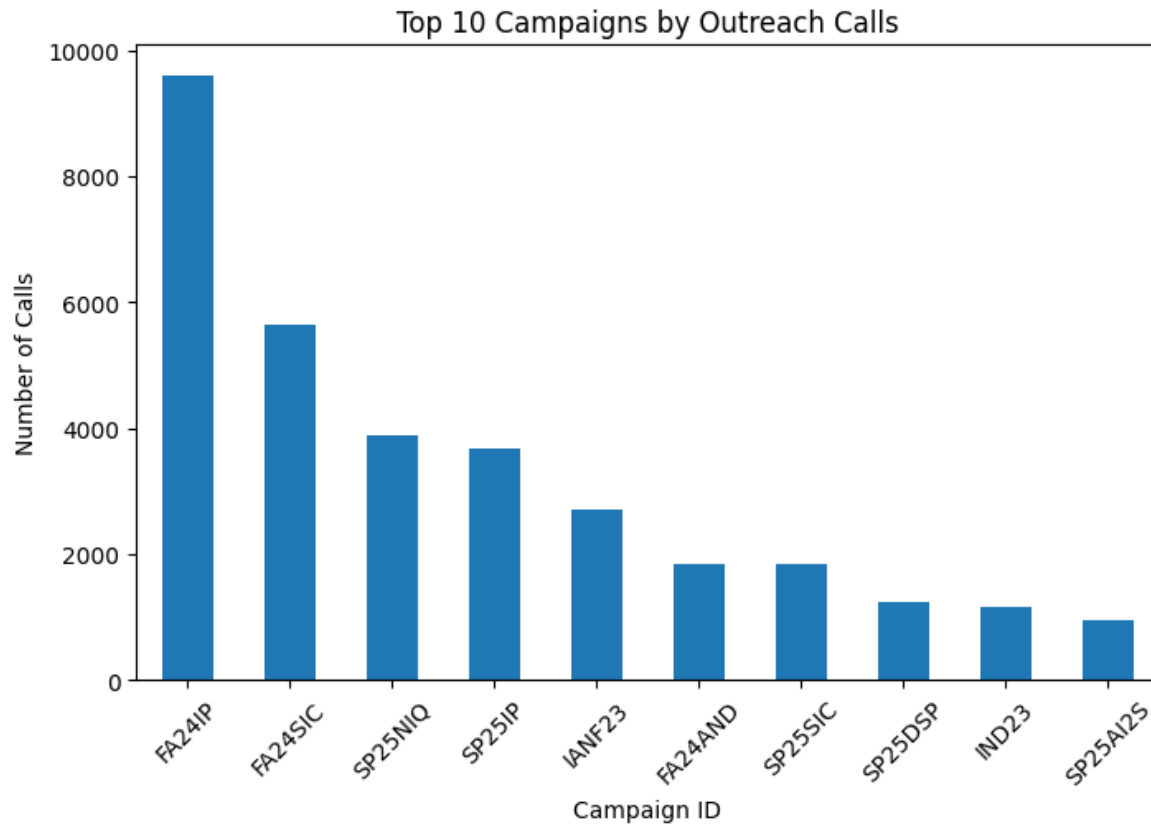


Campaign Launches Over Time (Line Chart)

Shows monthly distribution of campaigns. Clear spike in May 2023 (7 launches).

Calls per Campaign (Top 10)

Campaign_ID	Calls
FA24IP	9,603
FA24SIC	5,640
SP25NIQ	3,891
SP25IP	3,667
IANF23	2,697
FA24AND	1,846
SP25SIC	1,838
SP25DSP	1,229
IND23	1,168
SP25AI2S	948



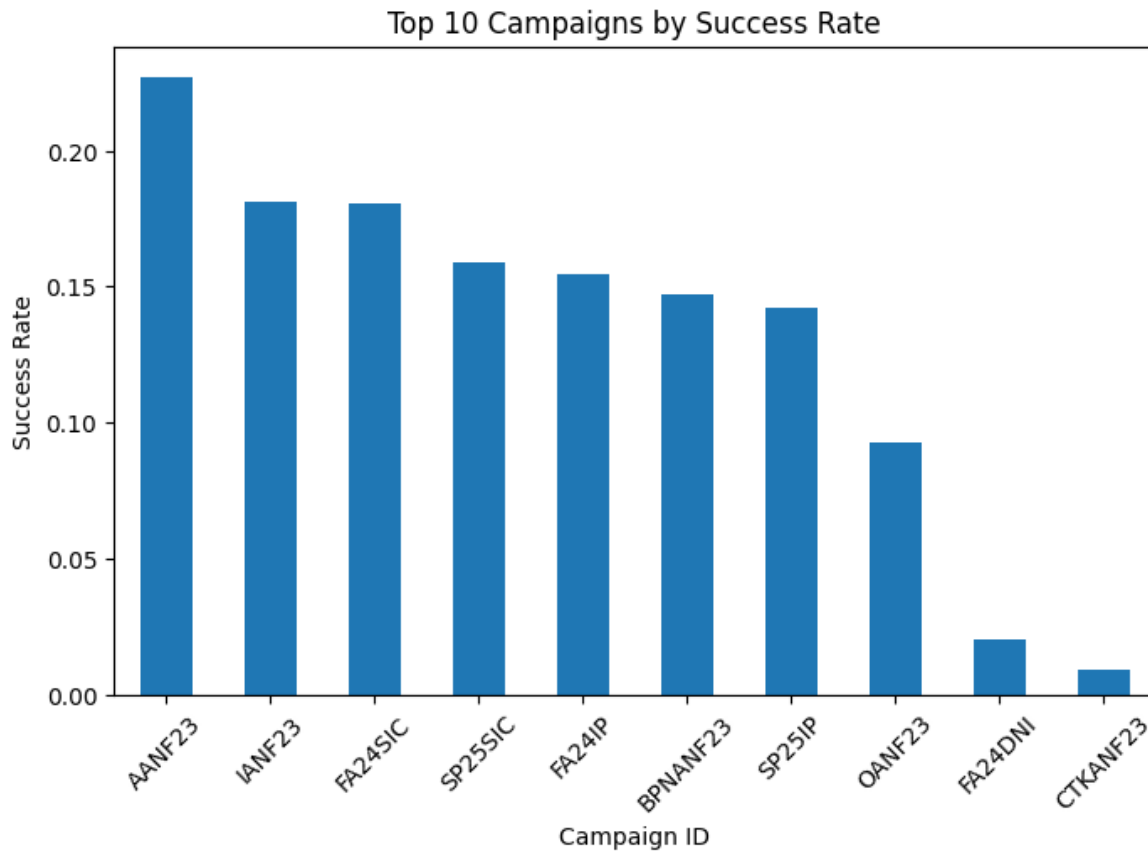
Calls per Campaign (Bar Chart)

FA24IP received the most outreach (9.6k calls), showing heavy resource allocation.

Success Rates (Top 10 Campaigns)

Campaign_ID	Success Rate
AANF23	22.7%
IANF23	18.1%
FA24SIC	18.0%
SP25SIC	15.9%
FA24IP	15.5%
BPANF23	14.7%
SP25IP	14.2%

OANF23	9.3%
FA24DNI	2.0%
CTKANF23	0.9%



Campaign Success Rates (Bar Chart)

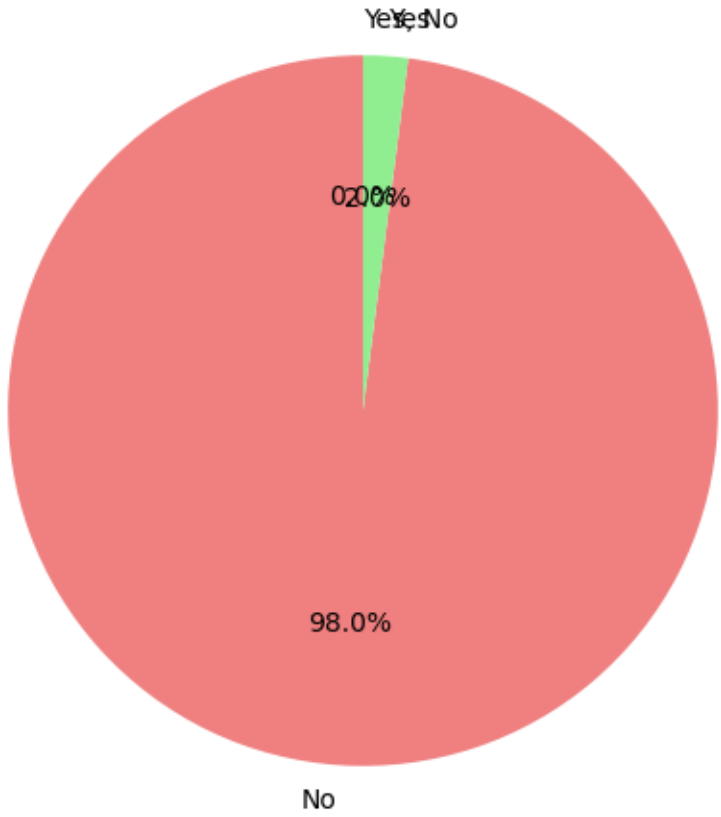
AANF23 & FA24SIC performed best; CTKANF23 had minimal conversions.

Escalations per Campaign

Campaign_ID	No (%)	Yes (%)	Both (%)
AANF23	97.8	2.2	0.0
AND23	93.4	6.6	0.0
CTKANF23	99.1	0.9	0.0

FA24AND	98.4	1.6	0.0
FA24DNI	98.4	1.6	0.0

Escalation Distribution Across Outreach Calls



Escalation Distribution (Pie Chart)

Majority of campaigns had <3% **escalations**, showing frontline resolution was strong.

Summary of Campaign Dataset

- **Campaign + Outreach Integration:** Linking with Campaign_ID provided campaign-level performance insights.
- **Campaign Activity:** Concentrated around **May 2023** and mid-2024.
- **Outreach Effort:** **FA24IP** received the highest call volume, while smaller campaigns had minimal outreach.
- **Effectiveness:** Conversion success rates vary widely (22% → <1%), highlighting inconsistency.
- **Escalations:** Rare (<5% for most), showing good query handling by outreach teams.

3 CONCLUSION

The exploratory data analysis (EDA) conducted on the Outreach, Applicant, and Campaign datasets provides an integrated view of how campaigns are designed, executed, and converted into applications. The findings reveal strong activity levels and resolution efficiency but also highlight gaps in consistency, targeting, and data quality.

Outreach activity showed high call volumes concentrated around campaign launch periods, with a small group of callers contributing disproportionately. While most queries were resolved without escalation (<3%), the effectiveness of calls varied, with some campaigns achieving strong success rates despite lower outreach, while others consumed significant effort but yielded weak conversions. This indicates that campaign design and targeting are more critical than pure outreach volume.

Applicant data revealed healthy inflows from certain regions, but missing contact details limited follow-up opportunities. Application surges near deadlines reflected alignment with outreach pushes, confirming that campaign timing influences applicant behavior. However, regional imbalance and incomplete data reduce overall conversion potential.

Campaign-level analysis confirmed wide variability: while campaigns like AANF23 and FA24SIC converted strongly (>18%), others like CTKANF23 saw almost no success despite outreach attempts. Linking campaign and outreach datasets (via Campaign_ID) made it clear that high-effort campaigns do not always deliver proportionate results, pointing to opportunities for refining targeting strategies.

Taken together, the analysis emphasizes three key points:

1. Quality over Quantity → More calls do not guarantee higher conversions; targeted campaigns matter more.
2. Operational Efficiency → Call handling is effective with minimal escalations, but caller performance and data completeness remain inconsistent.
3. Strategic Opportunity → Improving applicant data collection, diversifying regional outreach, and optimizing campaign targeting will significantly enhance outcomes.

Overall, the institution's outreach and campaign efforts are active and reasonably effective but not fully optimized. By focusing on data completeness, better campaign targeting, and consistent outreach practices, the organization can achieve higher ROI from its campaigns and build stronger, more sustainable applicant pipelines.